

Benedikt Seiter

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EDUCATION

ETH Zürich

Zurich, Switzerland

M.Sc. in Applied Mathematics, focus area: Machine Learning

Sep 2026 – Spring 2028 (expected)

- **Current coursework:** Probabilistic Artificial Intelligence, Natural Language Processing, Advanced Machine Learning

B.Sc. in Mathematics

Sep 2023 – Aug 2026

- **GPA:** 5.6 / 6.0
- **Thesis:** *Anchor PCA* (6.0 / 6.0), advisor: Prof. Jonas Peters; resulting paper accepted at NeurIPS 2026.
- **Selected coursework:** Mathematical Statistics, High-Dimensional Statistics, Computational Statistics, Machine Learning, Measure Theory, Probability Theory

RESEARCH & PROJECTS

Anchor PCA

Apr 2026 – Present

ETH Zürich, Seminar for Statistics

B. Seiter, A. Fries, J. von Kügelgen, and J. Peters | NeurIPS 2026 (accepted) | [arXiv:2606.06233](https://arxiv.org/abs/2606.06233)

- Developed Anchor PCA, an efficient method for multi-domain dimensionality reduction that balances explained variance with stability across shifting data regimes; implemented its estimators and supporting utilities in Python using NumPy and SciPy.
- Designed and ran an end-to-end experimental pipeline benchmarking Anchor PCA against pooled and worst-case PCA baselines; demonstrated improved out-of-domain explained variance on real-world financial and gas sensor datasets.
- Established recovery of a maximal invariant subspace and minimax robustness guarantees.

Benchmarking Invariant Risk Minimization for Financial Time Series

Sep 2025 – Jan 2026

ETH Zürich, Supervised Research Project

- Reproduced and extended InvariantStock, an IRM-based VAE for regime-robust equity return prediction, in Python using PyTorch; rebuilt its data pipeline and ran comparative benchmarks, VIX-based environment ablations, and loss-function tests.

Seminar on Representation Learning

Sep 2025 – Jan 2026

ETH Zürich

- Analyzed and presented papers on representation learning; topics spanned deep learning, autoencoders, ICA, contrastive/self-supervised learning, causality, and single-cell biology applications.

EXPERIENCE

ETH Zürich, Teaching Assistant

Sep 2025 – Present

- Lead weekly problem-solving sessions in Theoretical Computer Science and Probability & Statistics; grade problem sets and final examinations.

AWARDS

First Prize, Bundeswettbewerb Mathematik

2023

- Top 2.5% of participants in Germany's federal mathematics competition; invited to the German IMO team selection examinations.

DPG-Abitur- und Buchpreis (Physics)

2023

Dr. Hans Riegel-Fachpreis (Mathematics)

2022

SKILLS

Programming: Python, C++, R, MATLAB

Frameworks/Libraries: PyTorch, NumPy, SciPy, scikit-learn, pandas

Tools: Git, LaTeX | **Languages:** German (native), English (C2, IELTS 8.5)